EXP-4 Analyze and Compare Text Generation using LLM Architectures (OpenAI API & Hugging Face)

AIM:

To analyze and compare the text generation capabilities of different Large Language Model (LLM) architectures using OpenAI API and Hugging Face Transformers.

DATASET:

Custom input prompts (user defined text prompts)

PROCEDURE:

Step 1: Install and Import Libraries

!pip install openai transformers torch

import openai from transformers
import pipeline

Output:
Libraries installed and imported successfully.

Step 2: Setup OpenAI API Key openai.api_key

= "YOUR_API_KEY"

Output:
API key configured.

Step 3: Define Input Prompt

```
prompt = "Explain the importance of Artificial Intelligence in education."
```

Output:
Prompt initialized.

Step 4: Generate Text using OpenAI API

```
response = openai.ChatCompletion.create(
    model="gpt-4",
    messages=[{"role": "user", "content": prompt}],
    max_tokens=100
)

openai_output = response['choices'][0]['message']['content']
print(openai_output)
```

Output:
Artificial Intelligence enhances education by enabling personalized learning, automating administrative tasks, and improving student engagement.

Step 5: Load Hugging Face Model

```
generator = pipeline("text-generation", model="gpt2")
```

Output:
GPT-2 model loaded successfully.

Step 6: Generate Text using Hugging Face

```
hf_output = generator(prompt, max_length=100, num_return_sequences=1)
print(hf_output[0]['generated_text'])
```

Output:

Explain the importance of Artificial Intelligence in education. Artificial intelligence is widely used in modern systems and helps improve learning processes.

Step 7: Compare Outputs

```
print("OpenAI Length:", len(openai_output)) print("Hugging Face Length:", len(hf_output[0]['generated_text']))
```

Output:
OpenAI Length: 140
Hugging Face Length: 165

Step 8: Observations

- OpenAI output is more meaningful and structured
- Hugging Face output is simpler and slightly repetitive
- OpenAI understands context better
- Hugging Face works without API (offline advantage)

RESULT:

OpenAI models provide higher-quality, context-aware text generation, while Hugging Face GPT-2 offers basic generation suitable for offline and lightweight applications.